2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSC-1001

(Advanced Database Systems)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question no.*** **1**, ***question no.*** **2** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

   (a) What is a derived horizontal fragmentation? Give an example.

   (b) Define image of a database in a site for spatially distributed database system.

   (c) What is join graph?

   (d) Why name node does not become a bottleneck in achieving fast retrieval of data in HDFS?

   (e) How do you correlate Lattice as an algebraic structure with OLAP operations over multiple dimensions?

   (f) What is the purpose of Pivot operation on a cube data model?

   (g) Define OLAP.

2. Comment on the correctness of the following statements and justify your opinion in brief (within 200 words each). Answer ***any five*** questions : 4×5

   (a) It is useless for a hacker to get a dump of one or more data nodes in HDFS.

   (b) Isolation facilitates concurrent execution of transactions.

   (c) Relational data model may not be very efficient for a semi-structured database application.

   (d) Horizontal fragments are necessarily disjoint.

   (e) The availability of a distributed system degrades gracefully in the presence of an increasing number of faults.

   (f) Allocation schema is location-independent although it defines the allocation of fragments for a distributed database system.

   (g) Client applications invoked from computers at different locations, all connected to a remote database server with a back-up server, is an example of a distributed database.

Please Turn Over

3. (a) Describe the 6-layer architecture for a distributed database system and state the roles for each layer.

   (b) How semi-join operation significantly reduces the cost of query processing in a distributed environment? 6+4

4. (a) Define min-term predicate.

   (b) Explain using a suitable example the importance of a simple join graph towards designing a distributed database system.

   (c) While the allocation schema deals with allocation of fragments to different sites, explain why is allocation schema considered as a site-independent schema. 2+5+3

5. (a) "Vertical fragments are not disjoint." — Comment on the correctness of the statement and justify your opinion.

   (b) Evaluate the efficiency of using derived fragmentation for a spatial database using a suitable example. 5+5

6. (a) What would be the impact of different granularity of lock sizes for concurrency control in a distributed database environment?

   (b) Is the 2-phase commit protocol consistent with the ACID properties for transaction management? Explain with justification. 5+5

7. (a) What are the different types of filters used for Hyper file data model to extract data? State the basic differences between those with a suitable example.

   (b) Qiaos Kitchen is a fast food chain selling Japanese food. Different dishes sold require different number of ingredients in diverse proportion. The employee profile also varies depending on the types of employees and their job functionality.

      (i) Write a query to retrieve the employee records who works in the sales counters of Qiaos Kitchen.

      (ii) What would be the difference in output for the following two queries on Dishes?

         - Dishes | (string, type, "spicy") | [| (Pointer, FOT, ?X) | ↑↑ X]^3 | Target
         - Dishes | (string, type, "spicy") | [| (Pointer, FOT, ?X) | ↑ X]^3 | Target

      Justify your answer. 3+(3+4)

8. (a) Define the following terms in the context of HDFS :

      (i) Heartbeat (ii) Block Report (iii) Name Node (iv) Replication Factor.

   (b) Explain how in HADOOP, the replicas of a block can be created and stored in multiple data nodes efficiently.

   (c) What is Safe mode in Hadoop?

   (d) How Snow-Flake is different from a Star Schema for data warehouse? 4+3+2+1